

BioNode — Investor Prep & Technical Vulnerability Brief

BEAR Summit 2026 · CES Pavilion
Top 10
Team copy — confidential

1 The frame: tier every claim before you speak

Critical investors and CES judges punish blurred lines between what runs today and what is projected. Win by drawing the line yourself — loudly. Label every claim as one of three tiers. It signals engineering maturity and disarms the "is this real?" attack before it lands.

✓ PROVEN TODAY

You can verify it at our booth

System architecture & AIoT control design. Live Edge-AI dashboard catching a pH fault and auto-correcting. Bench-scale digestion producing measurable biogas. Finalised BOM + local sourcing.

⚙ BUILDING — BY NOV MVP

Under BEAR mentorship, on a costed plan

Closing the full chain to a filled 200-bar cylinder. Professional casing. First deployable, costed unit. Steady-state throughput numbers.

© VISION — DIRECTIONAL

Clearly flagged as ambition

\$44M ARR, 3-unit pilot, 10,000 units / 50 countries / 2036. Never stated as fact — always "our model targets..."

2 Technical vulnerabilities — and how to defuse each

These are the questions a sharp energy investor or process engineer **will** ask. Each is a genuine weakness in the current plan. Do not dodge — pre-empt. The credible answer is almost never "it's solved"; it is "here's the number, here's the risk, here's how the mentored build closes it."

① Energy balance — the parasitic-load question

HIGHEST RISK

The attack: "Compressing gas to 200 bar and heating a digester are both energy-hungry. Does one container produce more energy than it burns — or is your floating solar array quietly subsidising a net-negative machine?"

Your answer: Own the number. "Multi-stage compression to 200 bar costs roughly **0.3–0.4 kWh per kg** of gas; mesophilic digestion needs ~35–38 °C. Our energy model budgets both against the aquavoltaic array plus battery buffer, with grid as backup — the AI load-shifts to protect the balance. We treat net-energy-positive operation as a **design gate for the MVP**, not an assumption. The bench unit is where we measure the real coefficient, and we will publish it." Confidence comes from having the figure and calling it a gate — not from claiming it's already won.

Have ready: a one-line energy budget (kWh in vs. MJ of BioCNG out per cycle) and your PV array sizing assumption, even if provisional.

② Feedstock mass balance — hyacinth is 95% water

HIGHEST RISK

The attack: "Water hyacinth is mostly water and high in lignin — low biogas yield per tonne. To fill vehicle cylinders daily you need tonnes of it and a big digester. How is that 'container-sized'?"

Your answer: "Correct — fresh hyacinth needs chopping and partial dewatering, and yield per fresh tonne is modest. That's why the container is a **micro-refinery, not a mega-plant**: it serves one terminal's fill volume, not a city. Our bench unit measures actual m³ biogas per kg of prepared feedstock; that number sizes the digester and tells us the true daily output. We would rather quote a real yield from our own digester than an optimistic textbook figure." Naming the pretreatment step shows you understand the biology.

Have ready: target kg-feedstock → m³-biogas → kg-BioCNG per day, flagged as "to be confirmed on the bench unit."

③ Safety & regulation — 200-bar flammable gas, in public

HIGHEST RISK

The attack: "You're dispensing explosive gas at 200 bar to an untrained queue of drivers at a public terminal. Who certifies that? What's your H₂S and moisture spec? This is a licensing and liability minefield."

Your answer: "We treat compliance as core engineering, not paperwork. Vehicle-grade gas requires H₂S below the corrosion threshold and moisture removal — our scrubbing + separation targets that spec, and we test it. On-site dispensing falls under the **Dept. of Explosives and BSTI** in Bangladesh; part of the mentored phase is mapping that pathway with the terminal operator as partner. For the pilot we can also fill certified cylinders off-peak rather than open-dispense, which shrinks the regulatory surface while we build the track record." Show you know the two regulators by name.

Have ready: your target gas spec (CH₄ %, H₂S ppm, moisture) and the phrase "Dept. of Explosives + BSTI licensing pathway."

④ Unit economics — \$1,330 is a demo, not a revenue unit

HIGHEST RISK

The attack: "Your CAPEX is BDT 1,46,500 (~\$1,330), yet a single multi-stage 200-bar compressor costs more than that alone. What does a **deployable revenue unit** actually cost — and what's the payback?"

Your answer: "That figure is explicitly our **Phase-1 bench-demo BOM** — sensors, PLC, small digester — not the deployable unit. We're transparent that a revenue-grade unit with a rated compressor and certified vessel is a different capital tier. Building the honest per-unit

CAPEX, OPEX (feedstock labour, iron-sponge replacement, maintenance) and payback model is a **named deliverable of the mentored MVP phase**. We'd rather show you a real bench cost now and a real unit cost in November than a hand-waved number today." Never let the \$1,330 be mistaken for the unit price.

Have ready: *a clear split — "bench BOM ~\$1,330" vs. "deployable unit: costing under mentorship."*

⑤ "AI" precision — automation vs. machine learning

CREDIBILITY

The attack (from an AI-literate judge): "You call it Edge-AI and a 'machine-learning inference engine'. On one digester with no history, what is the model actually trained on? Isn't this really PID setpoint control?"

Your answer: "Today it's a **sensor-driven control loop with anomaly detection** — closed-loop pH/temperature regulation plus fault-catch logic, which is exactly what the live dashboard shows. The learning layer grows as units log data: the ML value is in **predicting a digester crash before it happens** and load-shifting energy across a fleet. We're precise about that distinction because over-claiming 'AI' to this room costs more credibility than it buys." Precision here reads as senior, not weak.

Have ready: *the sentence "control + anomaly detection now; predictive models as fleet data accumulates."*

⑥ Digestion timescale — 72-hour run ≠ daily output

PROCESS

The attack: "Anaerobic digestion has a 20–40 day retention time. A 72-hour run proves startup, not the steady daily fill you describe at Gabtoli. How do you get continuous output?"

Your answer: "Right — a batch digester has long retention. Continuous daily output comes from **semi-continuous feeding at steady state**: once the digester is established, you feed and draw daily against a resident bacterial population. The 72-hour autonomous run at the summit proves the **control system holds conditions**; steady-state throughput is what the pilot demonstrates over weeks. We separate those two claims deliberately."

Have ready: *"72-hr run = control proof; steady-state daily fill = pilot proof." Don't conflate them on stage.*

⑦ Team composition — three electrical engineers

EXECUTION

The attack: "This is a bio-chemical-mechanical high-pressure product, and your team is three EEs. Who owns the digester biology and the pressure-vessel engineering?"

Your answer: "Our core moat **is** the control system — the hardest, most defensible part, and squarely our expertise. We're clear-eyed that AD biology and pressure engineering need dedicated depth, which is exactly what the **Dholaikhal fabrication partners and a process advisor** bring, and what we'll formalise under BEAR mentorship. Naming the gap and the plan to fill it is stronger than pretending three people cover everything." Investors back teams that know their own holes.

Have ready: *names/roles of your fabrication partner and any process/chemical advisor you can cite.*

⑧ Business model — EaaS makes you a regulated fuel retailer

COMMERCIAL

The attack: "Own-operate-sell means you're selling road fuel to the public — weights & measures, fuel standards, explosives licensing. And carbon credits need years of verified MRV. Aren't these revenue lines aspirational?"

Your answer: "Near-term revenue is the fuel margin — 20% below imported CNG, recurring daily, with the terminal operator as licensed partner rather than us going solo. **Carbon is upside, not base case**: it needs a Verra/Gold-Standard methodology and MRV, so we model it as a Year-2+ layer, never in the core payback. We keep the aspirational lines clearly out of the numbers that matter for a first investment."

Have ready: *"base case = fuel margin; carbon = flagged upside." Keep them in separate columns.*

⑨ Feedstock paradox & seasonality

SUPPLY

The attack: "Hyacinth growth slows in winter, and if you succeed at clearing waterways you destroy your own feedstock. What happens to a unit when the canal is clean?"

Your answer: "Hyacinth regrows aggressively — nationally it's a chronic surplus, not a one-time stock, so a single terminal's draw never outpaces a canal. For seasonality and siting flexibility the digester is **feedstock-agnostic**: it also takes market waste, napier grass, and agri-residue, so a unit is never idle. We size intake to a terminal's fill volume, well below regrowth rate."

Have ready: *"co-feedstocks: market waste, napier grass, agri-residue" as the seasonality hedge.*

⑩ Prototype maturity — no physical unit yet

STAGE

The attack: "Your own form says 'working prototype: No.' The Top-10 teams that reach CES built the machine. Where's yours?"

Your answer: "That form was the April pre-proposal. Since selection we've moved from paper to bench: [state exactly what physically runs on summit day — digester + scrubbing + live dashboard]. We agree completely that **the machine is what matters** — which is why the mentored phase to November is entirely about closing the chain to a filled cylinder. We're a build team, and the booth shows the build." Update this line to your real summit-day status — never overstate it.

Have ready: *one honest sentence naming exactly what physically operates at the booth.*

BioNode — Rapid-Fire Q&A + Tips

Judges score: problem · differentiation · demo · impact · scalability · commercialization · vision · team

VISION & SCALE

Q Isn't "1 billion people, 10,000 units" pure fantasy?

"It's our 2036 vision — clearly flagged as directional, the same way solar became rural-electrification standard. What earns the right to say it is the near-term proof: one unit running at one terminal with real numbers. We separate the dream from the deliverable on purpose."

Q Why does this deserve a global CES stage, not just a local grant?

"The demo is universally legible — **weed in, vehicle fuel out** — and the problem isn't only ours: invasive aquatic biomass plus fuel-import dependency is shared by Vietnam, Nigeria, Myanmar, Pakistan. A solution born in Bangladesh that exports is exactly what CES showcases."

DEFENSIBILITY & COMPETITION

Q What stops a large energy company copying this in a year?

"The moat isn't the chemistry — it's the **autonomous control system tuned on real digester data**, plus first-mover terminal relationships and local low-cost fabrication. A multinational won't out-cheap a Dholaikhal-built, Nawabpur-sourced unit for this market, and every unit we run widens our data lead."

Q Why hasn't anyone done weed-to-vehicle-fuel already?

"Pieces exist — biogas, upgrading, compression — but always as separate, operator-heavy, imported, multi-crore systems. Nobody collapsed them into one autonomous, locally-built, point-of-use container. The integration and the AI autonomy are the novelty."

THE ASK & MONEY

Q What do you actually need right now?

"Not a large cheque yet — introductions. A terminal partner for the pilot site, a process/pressure advisor, and the regulatory pathway with Dept. of Explosives. Capital scales **after** the pilot shows real kg-sold numbers; a Series A before that is premature and we won't pretend otherwise."

Q When is this investable, and what's the milestone?

"The inflection is the **3-unit Dhaka pilot with field revenue** — kg of BioCNG sold, to how many drivers, over how many days. That converts every projection into evidence. Back us at the pilot and you're buying proof, not a pitch."

IMPACT HONESTY

Q Your CO₂ and carbon-credit numbers — real or decorative?

"Modelled, and flagged as such. The 15,000 t/yr offset is at 200-unit scale, not today. Carbon revenue needs verified MRV, so it sits in Year-2+ upside, never in the base payback. We'd rather under-promise the credits and over-deliver."

✓ Summit-day tips — how to win the room

- ▶ **Lead with the physical demo, not slides.** Weed in one hand, the live dashboard behind you catching a pH fault. CES-style drama beats any chart — your own CES memo says so.
- ▶ **Weaponise honesty.** This judging panel rewards "reliability & demonstration quality" and candor. Saying "here's what runs, here's what we're building" out-scores teams that oversell a finished product.
- ▶ **Put a number on every claim or drop it.** "≥95% CH₄", "0.3–0.4 kWh/kg", "20% below imported CNG". Vague adjectives invite the exact attacks in this brief.

- ▶ **Keep a physical artifact on the table.** A jar of hyacinth, a piece of iron sponge, the sensor board. Investors remember what they can touch.
- ▶ **Name the two killer risks before they do** — energy balance and safety/licensing. Pre-empting them signals a team that has thought past the hype.
- ▶ **Assign roles at the booth:** one on the live demo, one on business/regulatory Q&A, one pulling contacts into a follow-up list. Never let a serious investor leave without capturing them.
- ▶ **Close every serious conversation with the pilot ask** — terminal partner, advisor, or regulatory intro. Specific asks get specific help; "we're raising" gets nods and no action.